ETSI EN 300 959 V6.1.1 (2000-06)

European Standard (Telecommunications series)

Digital cellular telecommunications system (Phase 2+); Modulation (GSM 05.04 version 6.1.1 Release 1997)

Reference
REN/SMG-020504Q6R1

Keywords
Digital cellular telecommunications system,
Global System for Mobile communications (GSM)

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Sous-Préfecture de Grasse (06) N° 7803/88

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Contents

Intellectual Property Rights 4

Foreword 4

1 Scope 5
1.1 References 5
1.2 Abbreviations 5

2 Modulation format 5
2.1 Modulating bit rate 5
2.2 Start and stop of the burst 5
2.3 Differential encoding 6
2.4 Filtering 6
2.5 Output phase 7
2.6 Modulation 7

Annex A (informative): Change control history 8

History 9

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Foreword

This European Standard (Telecommunications series) has been produced by the Special Mobile Group (SMG).

The present document specifies the modulation format used within the digital cellular telecommunications system.

The contents of the present document are subject to continuing work within SMG and may change following formal SMG approval. Should SMG modify the contents of the present document it will then be republished by ETSI with an identifying change of release date and an increase in version number as follows:

Version 6.x.y

where:

6 indicates release 1997 of GSM Phase 2+

x the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.

y the third digit is incremented when editorial only changes have been incorporated in the specification.

National transposition dates	
Date of adoption of this EN:	12 May 2000
Date of latest announcement of this EN (doa):	31 August 2000
Date of latest publication of new National Standard or endorsement of this EN (dop/e):	28 February 2001
Date of withdrawal of any conflicting National Standard (dow):	28 February 2001

1 Scope

The modulator receives the bits from the encryption unit, see GSM 05.01 [1], and produces an RF signal. The filtering of the Radio Frequency (RF) signal necessary to obtain the spectral purity is not defined, neither are the tolerances associated with the theoretical filter requirements specified. These are contained in GSM 05.05 [4].

1.1 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies.
- A non-specific reference to an ETS shall also be taken to refer to later versions published as an EN with the same number.
- For this Release 1997 document, references to GSM documents are for Release 1997 versions (version 6.x.y).

[1] GSM 01.04: "Digital cellular telecommunication system (Phase 2+); Abbreviations and acronyms".

[2] GSM 05.01: "Digital cellular telecommunication system (Phase 2+); Physical layer on the radio path General description".

[3] GSM 05.02: "Digital cellular telecommunication system (Phase 2+); Multiplexing and multiple access on the radio path".

[4] GSM 05.05: "Digital cellular telecommunication system (Phase 2+); Radio transmission and reception".

1.2 Abbreviations

Abbreviations used in this specification are listed in GSM 01.04 [1].

2 Modulation format

2.1 Modulating bit rate

The modulating bit rate is $1/T$ = 1 625/6 kbit/s (i.e. approximately 270,833 kbit/s).

2.2 Start and stop of the burst

Before the first bit of the bursts as defined in GSM 05.02 [3] enters the modulator, the modulator has an internal state as if a modulating bit stream consisting of consecutive ones ($d_i = 1$) had entered the differential encoder. Also after the last bit of the time slot, the modulator has an internal state as if a modulating bit stream consisting of consecutive ones ($d_i = 1$) had continued to enter the differential encoder. These bits are called dummy bits and define the start and the stop of the active and the useful part of the burst as illustrated in figure 1. Nothing is specified about the actual phase of the modulator output signal outside the useful part of the burst.

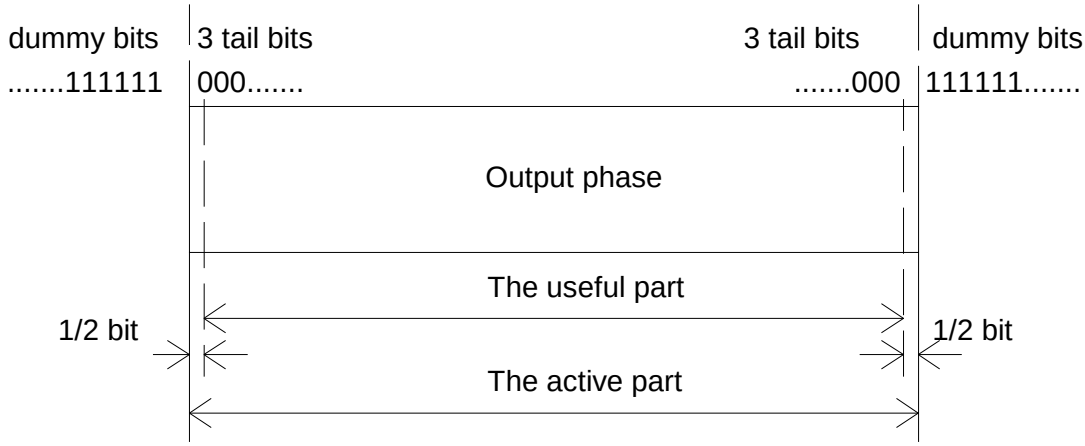

Figure 1: Relation between active part of burst, tail bits and dummy bits. For the normal burst the useful part lasts for 147 modulating bits

2.3 Differential encoding

Each data value $d_i = [0,1]$ is differentially encoded. The output of the differential encoder is:

$$\hat{d}_i = d_i \oplus d_{i-1} \qquad (d_i \in \{0,1\})$$

where $\oplus$ denotes modulo 2 addition.

The modulating data value α_i input to the modulator is:

$$\alpha_i = 1 - 2\hat{d}_i \qquad (\alpha_i \in \{-1,+1\})$$

2.4 Filtering

The modulating data values α_i as represented by Dirac pulses excite a linear filter with impulse response defined by:

$$g(t) = h(t) * rect\left(\frac{t}{T}\right)$$

where the function *rect(x)* is defined by:

$$rect\left(\frac{t}{T}\right) = \frac{1}{T} \qquad for\ |t| < \frac{T}{2}$$

$$rect\left(\frac{t}{T}\right) = 0 \qquad otherwise$$

and * means convolution. *h(t)* is defined by:

$$h(t) = \frac{\exp\left(\frac{-t^2}{2\delta^2 T^2}\right)}{\sqrt{(2\pi)}\ .\ \delta T}$$

where $\delta = \frac{\sqrt{\ln(2)}}{2\pi BT}$ *and* $BT = 0.3$

where B is the 3 dB bandwidth of the filter with impulse response $h(t)$, and T is the duration of one input data bit. This theoretical filter is associated with tolerances defined in GSM 05.05 [4].

2.5 Output phase

The phase of the modulated signal is:

$$\varphi(t') = \sum_i \alpha_i \pi h \int_{-\infty}^{t'-iT} g(u)du$$

where the modulating index h is 1/2 (maximum phase change in radians is π/2 per data interval).

The time reference $t' = 0$ is the start of the active part of the burst as shown in figure 1. This is also the start of the bit period of bit number 0 (the first tail bit) as defined in GSM 05.02 [2].

2.6 Modulation

The modulated RF carrier, except for start and stop of the TDMA burst may therefore be expressed as:

$$x(t') = \sqrt{\frac{2E_c}{T}} \cdot \cos(2\pi f_0 t' + \varphi(t') + \varphi_0)$$

where E_c is the energy per modulating bit, f_0 is the centre frequency and φ_0 is a random phase and is constant during one burst.

Annex A (informative): Change control history

SMG	SPEC	VERS	NEW_VE	PHA	SUBJECT
S27	05.04	5.0.1	6.0.0 (Ph.2)	R97	conversion to Release 97 EN
S27	05.04	6.0.0	6.0.1	R97	Version update for publication
S30b	05.04	6.0.1	6.1.0	R97	Correction of mistake for range alpha-sub-i in Clause 2.3
	05.04	6.1.0	6.1.1	R97	Upgrade to version 6.1.1 for publication

History

Document history			
V6.0.0	November 1998	One-step Approval Procedure	OAP 9910: 1998-11-06 to 1999-03-05
V6.0.1	March 1999	Publication	
V6.1.0	January 2000	One-step Approval Procedure	OAP 200019: 2000-01-12 to 2000-05-12
V6.1.1	June 2000	Publication	